1. two_sylow_search on fp_backend::Fp64: run records (started 20261017T0018Z)

Kind: benchmark runs, descriptive (interpretation: `../../report.pdf`).
Study: `research/two_sylow_fp64_20261017/`, protocol `PROTOCOL.md`, frozen at `e71198313` before the first run.
Command: `hwcost measure --fields fp64,w1 --two-sylow` per run (7 repetitions, about 100 ms per arm); every run exited 0 and completed every arm.

1.1. Runs and binaries

run	started / finished UTC	binary SHA-256 (first 16)	source
base-1	20261017T0018Z / 20261017T0019Z	`1b74e265fe929a4e`	protocol commit e71198313423b83ca939162e2a6827c9c624e955 with src/cryptanalysis/fp_backend.rs at f25d5e520
cand-1	20261017T0019Z / 20261017T0020Z	`a7be862e657745ed`	protocol commit e71198313423b83ca939162e2a6827c9c624e955
base-2	20261017T0020Z / 20261017T0021Z	`1b74e265fe929a4e`	protocol commit e71198313423b83ca939162e2a6827c9c624e955 with src/cryptanalysis/fp_backend.rs at f25d5e520
cand-2	20261017T0021Z / 20261017T0023Z	`a7be862e657745ed`	protocol commit e71198313423b83ca939162e2a6827c9c624e955

Host: Intel(R) Xeon(R) Processor @ 2.10GHz (signature 6/207/2), 4 vCPUs, cpu_flag = hypervisor, systemd_detect_virt_vm = kvm isolation L0; no PMU; no other build or test ran during the runs.

1.2. Gate (G)

Each run's `gate.txt` (the `two_sylow` lines of its `stderr.txt`):

```
two_sylow z2xz2 p=9223372036854775783: 2000 draws, three arms agree on every depth; depth sum 4922, points (depth >= 6) 85, x xor per arm baseline 0x4e343ade76751fd0, fast-chi 0x5fb70c5dbe2d5f61, fast 0x363b8fb6939ce2cb
two_sylow z2xz8 p=9223372036854775783: 2000 draws, three arms agree on every depth; depth sum 7939, points (depth >= 6) 236, x xor per arm baseline 0x27cf114294b01830, fast-chi 0x56b86e862fb28989, fast 0x49903f974568e2dc
```

`G PASS: gate lines identical in all four runs`

1.3. Claims (`hwcost compare`)

id	statement	value	threshold	verdict
S1	max(r1, r2) over 6 arms tsy.* (batch ≥ 0)	1.0832	1.1	PASS
K1	max(r1, r2) over 6 arms kernel.fp64.* (batch ≥ 4096)	0.4415	0.85	PASS
C1	max |r_k − 1| over 6 arms kernel.w1.* (batch ≥ 4096)	0.0445	0.1	PASS

1.4. two_sylow_search arms (ns per draw, medians of 7)

arm	A1	B1	A2	B2	r1	r2	drift	margin	class
tsy.z2xz2.baseline	6173	6251	6136	6510	1.013	1.061	0.006	0.02	no detected effect
tsy.z2xz2.fast	3594	3600	3596	3887	1.001	1.081	0	0.02	no detected effect
tsy.z2xz2.fast-chi	4314	4415	4279	4635	1.023	1.083	0.008	0.02	candidate slower
tsy.z2xz8.baseline	7052	6892	6804	6952	0.977	1.022	0.035	0.07	no detected effect
tsy.z2xz8.fast	2504	2474	2478	2562	0.988	1.034	0.011	0.021	no detected effect
tsy.z2xz8.fast-chi	3063	3111	3061	3276	1.016	1.07	0.001	0.02	no detected effect

Every sample (ns per draw, repetition order):

arm	base-1	cand-1	base-2	cand-2
tsy.z2xz2.baseline	6369, 6197, 6128, 6173, 6118, 6432, 6125	6413, 6072, 6236, 6251, 6260, 6318, 6084	7509, 6073, 6101, 6136, 6116, 6173, 6155	6090, 6510, 7672, 7143, 6167, 6728, 6179
tsy.z2xz2.fast	3644, 4235, 3594, 3581, 3574, 3644, 3589	3539, 3639, 3582, 3669, 3699, 3600, 3586	3615, 3569, 3572, 3624, 3611, 3587, 3596	3530, 4168, 3887, 3710, 3622, 4076, 4109
tsy.z2xz2.fast-chi	4348, 4468, 4399, 4314, 4292, 4306, 4302	4828, 5077, 4411, 4415, 4588, 4320, 4298	4390, 4273, 4253, 4291, 4247, 4279, 4333	4278, 4681, 4454, 5118, 4635, 4463, 5155
tsy.z2xz8.baseline	7053, 7052, 6904, 6843, 7167, 7074, 6780	6885, 6892, 6853, 7065, 7057, 7097, 6805	6931, 6800, 6922, 6804, 6845, 6789, 6801	6769, 8141, 6988, 6952, 6839, 6951, 7414
tsy.z2xz8.fast	2713, 2504, 2498, 2472, 2479, 2700, 2537	2767, 2455, 2496, 2464, 2474, 2439, 2493	2835, 2478, 2468, 2472, 2480, 2487, 2476	2493, 2894, 2807, 2878, 2469, 2540, 2562
tsy.z2xz8.fast-chi	3151, 3097, 3058, 3063, 3058, 3080, 3055	4010, 3116, 3131, 3105, 3093, 3111, 3070	3338, 3048, 3031, 3061, 3079, 3068, 3035	3096, 3533, 3276, 3433, 3105, 3113, 3482

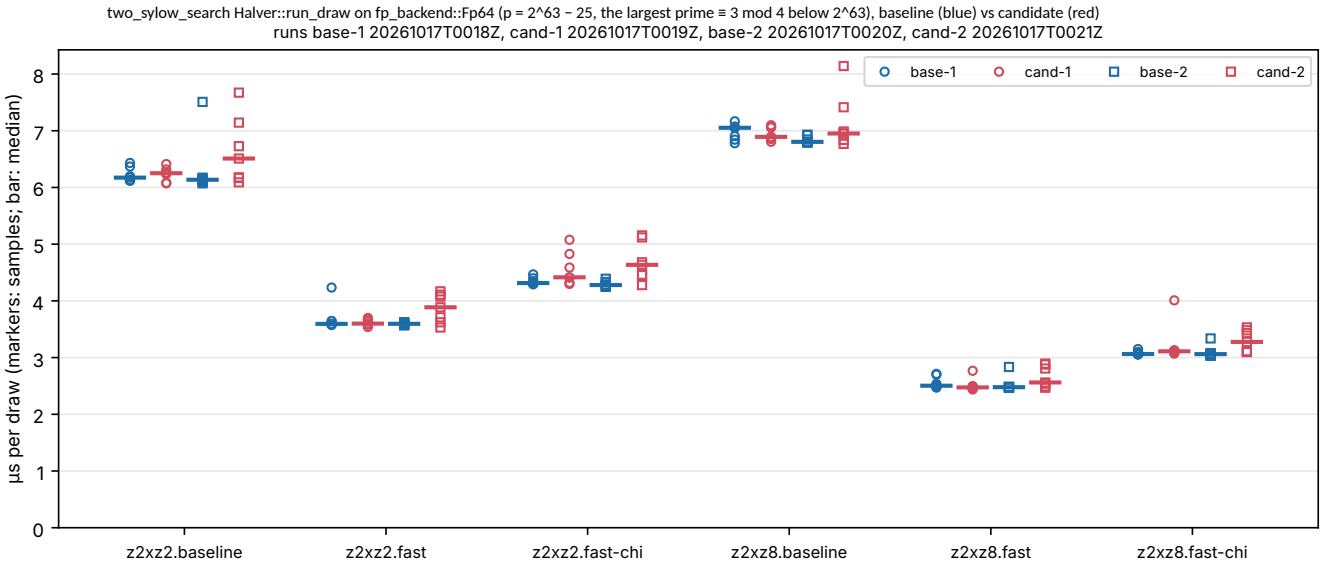


Figure 1: Every `tsy` sample per run.

1.5. Controls and other arms

Arm classes (effect called only beyond max(2·drift, 2%) on the same side in both pairs):

group	class	arms
clock	no detected effect	2
fp64	candidate slower	1
fp64	no detected effect	7
kernel	candidate faster	10
kernel	no detected effect	10
mem	candidate faster	1
mem	candidate slower	4
mem	no detected effect	15
scaling	candidate faster	3
scaling	no detected effect	3
tsy	candidate slower	1
tsy	no detected effect	5
w1	no detected effect	8

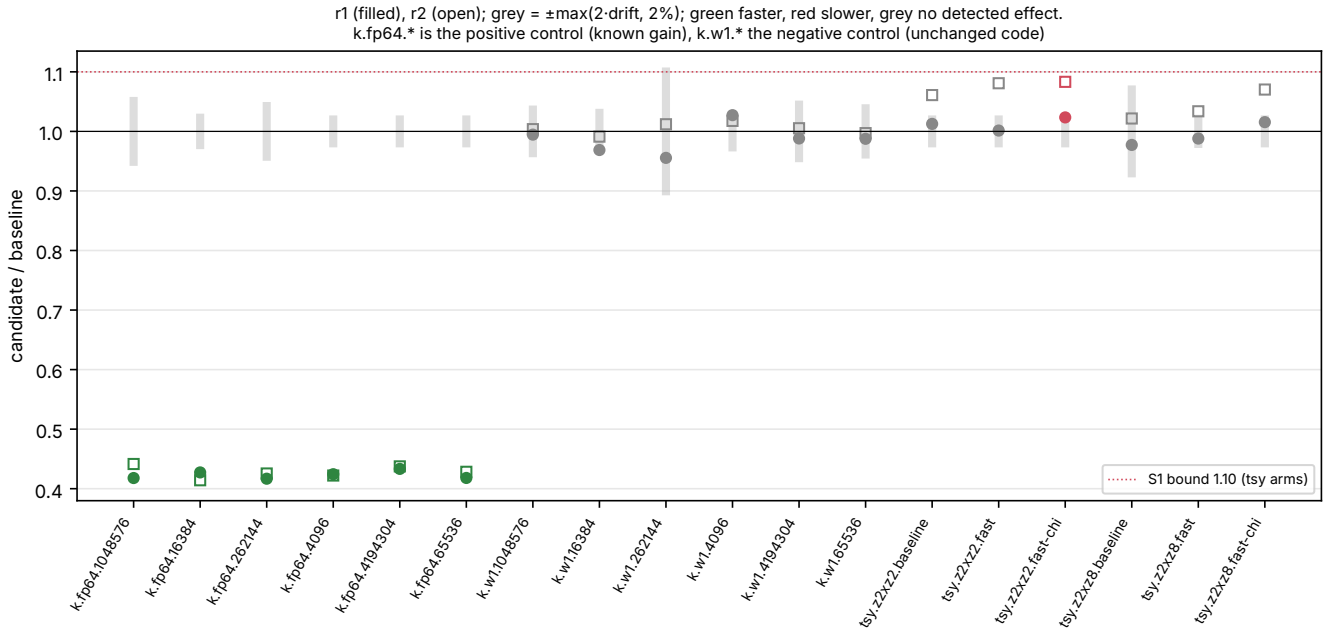


Figure 2: Ratios of the `tsy` arms and of the kernel controls at B ≥ 4096.

1.6. Raw records

`runs/{base,cand}-{1,2}/` (`measurements.json`, `samples.csv`, `medians.csv`, `stderr.txt`, `gate.txt`, `binary.txt`, timestamps), `runs/compare/`, `runs/protocol_commit.txt`. Graph source: `../../plot.py` (layout only).